

Anna-Drishti — Team Guide

Everyone except the lab. What you own, when it is due, and what you will be asked.

Chiranjib · Mohit · Vishal · Devavrath — SIH 2026 · VIT · issued 22 August

WHERE WE ACTUALLY ARE

The software is built and working end to end. Phases P1 to P5 of the plan are done — physics, CO₂, virtual sensors, decision engine, comparison mode. **P4, the milestone the whole weekend hinges on, is behind us.** We are ahead of schedule.

What remains is almost entirely unglamorous finishing: polish, rehearsal, the backup video, and each of us being able to defend the part we own. Those are the things that actually decide Monday, and they are the things teams usually skip.

THE RULE THAT MATTERS MOST

When a judge asks where a number came from, **the person who owns it answers — not the person who typed it.** A lot of numbers were typed this weekend by one person. They are not his. Read the part that belongs to you, correct anything wrong, and be ready to defend it. If you cannot defend it, we cut it rather than bluff.

SCHEDULE

WHEN	WHAT	WHO
Sat – today	Jars sealed. Sensor sourcing decided.	Krittika, Devavrath
Sun 12:00	Verification sheets returned. All constants signed off.	Everyone
Sun 14:00	FEATURE FREEZE. No new functionality, no exceptions.	Everyone
Sun afternoon	Polish, three clean runs, deck complete, roles assigned	Chiranjib + all
Sun evening	Two timed rehearsals. Backup video. Offline test. Q&A drill.	Everyone
Mon morning	Arrive early, silent run, present	Everyone

14:00 SUNDAY IS NOT NEGOTIABLE

Anything not working at 14:00 gets cut and is never mentioned on stage. The urge to add one more thing on Sunday evening is the single most common way a good demo dies on Monday morning. If someone proposes wiring live sensor data into the software on Sunday, the answer is no — that is exactly what the freeze exists to refuse.

FIRST THING, ALL OF YOU

Clone the repo, open `index.html` in a browser. **The pill in the header must go green.** Then press **Deep interior pocket** and watch it run. Do this before reading anything else — everyone needs to have seen it move before Monday.

`github.com/Chiranjib-x/anna-drishti`

Chiranjib — Integration, and the things nobody else can do

Track A · You are merge authority and on the critical path. Do not take on hardware work.

SUNDAY AFTERNOON — P6

1 Call the freeze at 14:00

Out loud, to everyone. Write down anything that does not work so the narrator knows what not to touch on stage.

2 Projector-distance check

Fonts, colours and the charts have to be readable from the back of a room, not from your desk. Test at actual projector resolution.

3 Three consecutive clean runs

Without refreshing the page. Watch for anything that leaks, slows, or gets stuck. This is the P6 exit test.

4 Assign roles

One narrator, one driver on the laptop. The other four answer only in their own domain. Six people talking over each other reads as chaos.

Narrator: Driver:

SUNDAY EVENING — P7, THE INSURANCE

5 Record the backup video

This is never cut, under any circumstances. A full screen capture of the complete demo working. If anything fails on Monday you play the video and keep talking.

6 Offline test on the real laptop

Wifi off, the browser you will actually use, projector resolution. The file has zero external references so it should pass — but run it on the actual machine anyway. That is the whole point of the test.

7 Two full timed rehearsals

Start to finish, including the walk to the laptop and the opening line. If you are over time, cut content now rather than live.

8 Pack

Laptop, charger, **HDMI and VGA adapters**, the HTML on a pendrive and emailed to all six, the backup video, the printed science one-pager, the proposal PDF.

MONDAY

- Arrive early. Test the projector **before** judges start circulating
- One silent full run to confirm nothing broke overnight
- Open with the hook, not the architecture
- **Let the judge inject the hotspot themselves.** Hands on the mouse beats any slide
- Land the comparison timeline, then stop talking and let them look at it
- State the simulation boundary plainly, then the roadmap and the ₹20,000 budget

☐ Freeze called ☐ 3 clean runs ☐ Roles assigned ☐ **Video recorded** ☐ Offline test passed ☐ Packed

Mohit & Vishal — Defend the engine

Tracks B and C · The code in your domains is a first draft. Own it and correct it.

READ THIS HONESTLY

The physics and detector code was written fast, by one person, to get the loop working. It is not finished work and it is not anyone's private property. Your job now is to read your section, find what is wrong, and be the person who can explain it. If you arrive Monday without having read it, we have three people who can answer technical questions and two who cannot — worse than if you had written it yourselves.

WHERE YOUR CODE IS

One file, `index.html`. Every section opens with a banner comment. Search for the banner text rather than trusting line numbers — they drift as we edit.

SEARCH FOR	LINES	OWNER	CONTAINS
PHYSICS (Track B)	254–316	Mohit	The solver, stability rule, respiration coupling, injection
SENSING + DECISION (Track C)	317–407	Vishal	Noise model, anomaly detection, <code>decide()</code>

MOHIT — THREE CLAIMS TO BE ABLE TO DEFEND

- M1

The stability condition

The solver is explicit finite-difference with zero-flux boundaries. It is only stable while $D \cdot dt/dx^2 \leq 0.25$ — currently 0.18. Understand why the limit exists and what breaks past it. **If anyone ever sees NaN on screen, this is the first thing to check.**
- M2

The coupling

A single respiration term drives **both** CO₂ production and heat release. They are not independent effects — they are the same biology observed two ways. This is the core of the model.
- M3

Check the cap

Respiration is capped at 4,000 mg/kg/day to stop runaway self-heating. Verify it is never reached during a normal run. If it is, the cap is doing physics rather than protecting us from it, and that is a problem.

☐ Checked ☐ Issue found:

VISHAL — FOUR CLAIMS, TWO OF THEM RECENTLY CHANGED

- V1

The noise model

Every reading gets Gaussian noise and quantization: CO₂ ±25 ppm + 1.5 %, RH ±1.0 %, temperature ±0.15 °C. **Check these against real datasheets** for the parts in our budget. “Our detector already runs on noisy readings, here is the noise model” is a far stronger answer than silence.
- V2

Depth-matched comparison — changed

CO₂ is compared **across lances at the same depth**, not against all nine ports. Depth stratification is real and about 1,000 ppm; comparing all nine buried the signal completely. **This is the change most likely to be questioned.**

V3 Alert threshold — changed

Fires at **3.0 σ sustained over 3 consecutive readings**, down from 4.0. Measured separation: clean grain peaks at 1.5 σ , a real hotspot reaches 4.2 σ . **Verify this yourself** — run the demo on clean grain for 60 days and confirm it stays silent.

☐ Verified silent ☐ False alarm seen:

V4 Risk weighting

35 % water activity, 40 % urgency, 25 % CO₂ anomaly, alerting at 45 out of 100. Sensible, or should it be weighted differently?

SUNDAY — BUG TRIAGE ONLY

After the freeze you two are on triage and nothing else. Every crash, every NaN, every stuck state. Write down anything you cannot fix so the narrator knows to avoid it.

YOUR Q&A QUESTIONS

ASKED OF	QUESTION
Mohit	“Why should I believe your diffusion model?”
Mohit	“What happens if your solver goes unstable?”
Vishal	“Why add noise to a simulation at all?”
Vishal	“Why a differential baseline instead of an absolute threshold?”

OPTIONAL, AND ONLY IF EVERYTHING ELSE IS DONE

Mohit — the insect module (logistic growth seeded at a focus, acoustic amplitude proportional to population) is a bonus that was never built. Physics is stable so it is genuinely available. But the plan is explicit: **the loop is never risked for a bonus**. Do not start it before the triage is clear.

Devavrath — Deployment, sensors and the coverage finding

Track F · You own the physical numbers and the only genuine engineering result we can produce this weekend.

NOTHING ORDERED ONLINE ARRIVES BEFORE MONDAY

Delivery to Vellore is two to four days. Everything in the buy list below is **nationals hardware** — buy it once, buy it right, but do not expect it for the internal round.

Today, check the VIT labs instead: Biotech, Instrumentation, the maker space, ECE final-year project cupboards. Ask for any CO₂ sensor at all, an SHT31 or BME280, and an ESP32 or any Arduino.

Borrowed parts are the only ones that can serve Monday.

1 · SIGN OFF THE PHYSICAL CONSTANTS — BY SUNDAY 12:00

Verification Sheet pages 6–8, twelve items: stack dimensions, cell size, the three diffusivities, bulk density, porosity, specific heat, ambient conditions, probe geometry.

THE QUESTION YOU WILL BE ASKED

“Is your moisture figure a molecular diffusivity?” — **No**. It is an *effective* coefficient lumping convective moisture migration together with diffusion, because in a real bag stack convection dominates. Molecular diffusion alone would be far slower. Say this in your own words.

2 · CARBON DIOXIDE SENSOR — THE DECISION THAT MATTERS

SENSOR	RANGE	BUS	ESP32	≈ ₹	VERDICT
Sensirion SCD30	0–40,000 ppm	I ² C	3.3 V native	5,000–6,500	Buy this
Sensirion SCD41	0–40,000 (spec'd to 5,000)	I ² C	3.3 V native	3,500–5,000	Good, less headroom
Winsen MH-Z19B	0–5,000 ppm	UART	3.3 V logic 5 V power	2,500–3,500	Saturates in grain

Get the SCD30. A sealed jar reads tens of thousands of ppm and a genuinely spoiling stack interior does too. A 5,000 ppm ceiling pegs and tells you nothing. It also returns humidity and temperature on the same chip, cutting part count.

DISABLE AUTOMATIC BASELINE CORRECTION — THIS WILL COST YOU A DAY

Every NDIR sensor assumes it sees fresh 400 ppm air every 24 hours and silently re-zeros to the lowest value it has seen. Inside grain it **never** sees 400 ppm, so ABC quietly recalibrates the elevated reading back down to “normal” and erases the exact signal we are measuring. Turn ABC off in firmware. **This is the single most common way CO₂ grain-monitoring projects fail.**

3 · HUMIDITY AND TEMPERATURE — THIS IS OUR WATER ACTIVITY

SENSOR	RH ACCURACY	BUS	≈ ₹	VERDICT
Sensirion SHT31-D	±2 %	I ² C 3.3 V	700–1,200	Buy three
SHT35 / SHT85	±1.5 %	I ² C 3.3 V	2,000–3,500	Better if budget allows

SENSOR	RH ACCURACY	BUS	≈ ₹	VERDICT
BME280	±3 %	I ² C 3.3 V	350–500	Too coarse for a _w
DHT22 / AM2302	±2–5 %	1-wire	250	Do not use

Not DHT22 — it drifts, it is slow, and it fails in exactly the condensing high-humidity conditions our own proposal names as a known failure mode. The SHT31 carries an **onboard heater** to burn off condensation, which is the mitigation we already promised in writing.

Accuracy matters more here than anywhere else: **relative humidity is water activity**. ±2 % RH becomes ±0.02 a_w, and we distinguish 0.82 from 0.85.

THE PART EVERYONE FORGETS — I²C ADDRESS COLLISION

The SHT31 has only **two** possible addresses (0x44 and 0x45, set by the ADDR pin). We want three, at three depths. **Three on one bus collide and it will not work.**

Fix: a **TCA9548A I²C multiplexer, about ₹250**. Eight channels, solves it completely, scales straight to nine sensors for the full three-lance array. Buy it now or rediscover this at two in the morning.

4 · CONTROLLER AND SUPPORT PARTS

ITEM	QTY	≈ ₹	WHY
ESP32-DevKitC (WROOM-32)	2	450–700 ea	Buy two. They die.
TCA9548A multiplexer	1	250	Non-optional
5 V 2 A supply	1	300	SCD30 and WiFi peak together
2.2 kΩ resistors	10	20	I ² C pull-ups for long runs
Shielded 4-core cable	5 m	400	Lance runs
Dupont and JST connectors	—	300	

Rough total ₹9,000–11,000, comfortably inside our ₹20,000 ceiling. Verify prices on Robu.in and Amazon.in before ordering — they drift.

5 · COMPATIBILITY — READ BEFORE WIRING

EVERYTHING ABOVE IS 3.3 V I²C, SO NO LEVEL SHIFTERS ARE NEEDED

This is deliberate. **ESP32 GPIO is not 5 V tolerant**, and mixing 5 V parts onto it is how boards get killed. An all-3.3 V I²C set removes that entire class of problem.

One exception, the MH-Z19B if you end up with one: **power it from 5 V, but its UART lines are 3.3 V**, so TX and RX connect to the ESP32 directly. Do not run it on 3.3 V power — the infrared lamp will not drive properly and readings drift low.

Second gotcha: the **SCD30 stretches the I²C clock up to 150 ms**. ESP32 handles it, but you may need to raise `Wire.setTimeout()` or reads return garbage.

I²C OVER 2 M IS OUT OF SPECIFICATION

Probe lances put sensors metres from the controller; I²C is designed for on-board distances. In order: drop the bus to 100 kHz, use 2.2 k Ω pull-ups instead of 4.7 k Ω , use shielded twisted cable. If it is still unreliable at full length, the real fix is a small microcontroller at each depth talking RS-485 — but do not design for that until the problem is measured.

6 · BRING-UP ORDER — DO NOT WIRE EVERYTHING AT ONCE

- 1 ESP32 alone. Blink an LED.
- 2 **One** SHT31 wired directly. Bus scan, confirm 0x44.
- 3 SCD30 alone. Confirm ~400–600 ppm in room air, **then disable ABC**.
- 4 Add the TCA9548A. Move the working SHT31 behind it.
- 5 Add sensors two and three on separate channels.
- 6 Extend to full cable length **last** — this is where signal integrity breaks.

7 · CALIBRATE AGAINST SALT STANDARDS

Saturated salt solutions hold a known water activity in the air above them. Standard method, salts already in the lab, costs essentially nothing.

SALT (SATURATED, 25 °C)	A _w	USE
Magnesium chloride	0.328	Dry end
Sodium chloride	0.753	Below our threshold
Potassium chloride	0.843	Straddles the 0.82 floor
Potassium sulphate	0.973	Above

This converts “our sensor says 0.87” into “our sensor is calibrated against standards and says 0.87.” Ask Krittika — these are her reagents.

8 · THE COVERAGE ANALYSIS — YOUR GENUINE FINDING

Nobody else can produce this and it needs no hardware at all. Open the demo, inject a hotspot at many different positions, record how many days each takes to be flagged. Produce one sentence:

“Three lances at this spacing detect N % of hotspots within X days.”

Then turn it into one clean chart for the deck. That answers the deployment question better than any amount of CAD.

HOTSPOT POSITION	DAY FLAGGED	HOTSPOT POSITION	DAY FLAGGED

9 · DEPLOYMENT SLIDE

Probe geometry, the coverage result, what the hardware would actually be, and the ₹20,000 budget from the submitted proposal.

10 · WHAT NOT TO BUILD

SKIP	WHY
Rotary manifold, pump, tubing	Multi-day mechanical build with servo and tubing failure modes. Adds nothing to the argument yet.
Acoustic insect detection	Needs insects we do not have, a trained classifier, and a quiet room.
Probe lance fabrication	Machining time for no rhetorical gain — a sensor in a jar makes the same point.
Wireless dashboard	Nothing needs it, and COM-port roulette on an unfamiliar laptop is a bad way to lose.

HARDWARE MUST NOT TOUCH INDEX.HTML

The software demo runs identically whether any sensor exists or not. The **Bench reading** panel takes typed numbers precisely so a dead sensor, a missing driver or a bad COM port cannot take the software down with it. Read the jar, type the number, done.

☐ Labs checked ☐ Pages 6–8 signed off ☐ Coverage data collected ☐ Chart made ☐ Deployment slide done

Signed: Date: